Stochastic Integration: Intuition, Construction, and Applications

References

Main references

- S. Shreve, *Stochastic Calculus for Finance II*, Chapter 4.
- I. Karatzas and S. Shreve, *Brownian Motion and Stochastic Calculus*, Chapter 2.
- T. Björk, *Arbitrage Theory in Continuous Time*, chapter on stochastic integration.

Scope of this lecture

We build the Itô integral just far enough to use it confidently. The Itô formula comes in the next lecture.

Roadmap

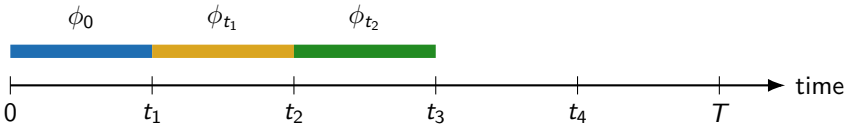
1. Start from discrete trading gains and understand the left-point rule.
2. Define the integral for simple predictable processes.
3. Use the Itô isometry to extend the definition.
4. Check the integrability needed for a true martingale.
5. Compute distributions and conditional laws in concrete examples.
6. Compare left-, midpoint-, and right-point conventions.
7. End with simulation rules and diagnostic checks.

Why Brownian Motion Needs a New Integral

A trading gain is a weighted sum of fresh price moves

On a grid $0 = t_0 < t_1 < \cdots < t_n = T$, let ϕ_{t_i} be the number of shares held during $(t_i, t_{i+1}]$. The cumulative gain is

$$G_T^{\pi} = \sum_{i=0}^{n-1} \phi_{t_i} \left(S_{t_{i+1}} - S_{t_i} \right).$$



Continuous-time target

The stochastic integral should be the limit of these gains as the trading mesh becomes finer.

Classical pathwise integration relies on smoothness or finite variation

For differentiable paths, one may write

$$\int_0^T \phi_t \, \mathrm{d}S_t = \int_0^T \phi_t S_t' \, \mathrm{d}t.$$

More generally, the Riemann–Stieltjes integral works well when the integrator has finite variation.

Smooth path

Small increments are essentially linear in Δt, so their total variation remains controlled.

Brownian path

ΔW has typical size $\sqrt{\Delta t}$. Total variation explodes as the mesh is refined.

Brownian motion is continuous, but it is almost surely nowhere differentiable and of infinite variation.

Quadratic variation is the scale that survives refinement

For a partition π of $[0, T]$, define

$$Q_T^\pi = \sum_i \left(W_{t_{i+1}} - W_{t_i}\right)^2.$$

If the grid has n equal intervals, each square is typically of size T/n, so the sum remains of order T.

Brownian quadratic variation

As $|\pi| \to 0$,

$$Q_T^\pi \longrightarrow T$$

in probability, and along standard nested partitions almost surely.

This surviving second-order term is why ordinary calculus must be modified.

The evaluation point changes the limiting answer

Take the integrand $\phi_t = W_t$. On the same partition, compare

$$L^\pi = \sum_i W_{t_i} \Delta W_i, \qquad R^\pi = \sum_i W_{t_{i+1}} \Delta W_i.$$

Since $W_{t_{i+1}} = W_{t_i} + \Delta W_i$,

$$R^\pi - L^\pi = \sum_i (\Delta W_i)^2 \longrightarrow T.$$

Consequence

There is no convention-free object called $\int W \, \mathrm{d}W$. The choice of sampling point matters by an order-one amount.

The $\mathrm{It\widehat{o}}$ integral uses left endpoints.

Left endpoints are an information constraint, not a cosmetic choice

At time t_i, the trader knows $\mathcal{F}_{t_i}$ and chooses ϕ_{t_i} before observing ΔS_i.

$$\underbrace{\phi_{t_i}}_{\mathcal{F}_{t_i}\text{-measurable}} \times \underbrace{(S_{t_{i+1}} - S_{t_i})}_{\text{future move}}.$$

Predictable in the discrete picture

The position used over $(t_i, t_{i+1}]$ must be determined by information available at t_i.

Using $\phi_{t_{i+1}}$ would allow the strategy to react after seeing the move it is supposed to earn.

Financial reading

The Itô convention encodes non-anticipative trading.

Quick exercise: which trading rules are admissible?

On the grid $0, 1, 2$, let $\Delta W_i = W_{i+1} - W_i$. Which positions are chosen without looking ahead?

1. $\phi_0 = 1$, $\phi_1 = 1 + W_1^2$.
2. $\phi_0 = \text{sign}(W_1)$, $\phi_1 = 0$.
3. $\phi_0 = 0$, $\phi_1 = \mathbf{1}_{\{W_1 > 0\}}$.
4. $\phi_i = \text{sign}(\Delta W_i)$.

Test

For the position held on $(i, i+1]$, ask whether it is $\mathcal{F}_i$-measurable.

Exercise debrief: admissibility is checked one interval at a time

Rule	Verdict	Reason
1	admissible	$1 + W_1^2$ is known at time 1
2	not admissible	W_1 is unknown at time 0
3	admissible	the event $\{W_1 > 0\}$ is in $\mathcal{F}_1$
4	not admissible	ΔW_i is the future move

Takeaway

An integrand may be random and path-dependent. What matters is that it uses only the past and present, never the next increment.

A predictable position cannot manufacture average drift

Suppose $S = W$ for the moment and $\phi_{t_i} \in \mathcal{F}_{t_i}$. Then

$$\mathbb{E}\big[\phi_{t_i} \Delta W_i \mid \mathcal{F}_{t_i}\big] = \phi_{t_i} \mathbb{E}[\Delta W_i \mid \mathcal{F}_{t_i}] = 0.$$

Summing and taking expectations gives

$$\mathbb{E}[G_T^\pi] = 0.$$

Sanity check

A non-anticipative strategy may change risk dramatically, but it cannot create a positive expected gain from a Brownian martingale.

This discrete identity will survive in the continuous-time integral.

Build the Integral from Simple Strategies

We integrate predictable processes with finite energy

Let $(W_t)_{0 \le t \le T}$ be Brownian motion with filtration $(\mathcal{F}_t)$. The intended integrand ϕ must satisfy two requirements:

1. **No anticipation:** ϕ_t is predictable.
2. **Finite energy:**

$$\mathbb{E} \int_0^T \phi_t^2 \, \mathrm{d}t < \infty.$$

Why the square?

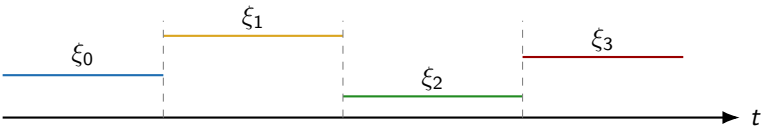
Brownian increments have variance Δt. This L^2 condition controls the gain variance and later guarantees the L^1 integrability required in the definition of a martingale.

We first define the integral for strategies that are constant between trading dates.

Simple predictable processes are continuous-time trading rules

Fix $0 = t_0 < t_1 < \cdots < t_n = T$. A simple predictable process has the form

$$\phi_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{(t_i, t_{i+1}]}(t), \qquad \xi_i \in \mathcal{F}_{t_i}.$$



The random coefficient ξ_i is chosen at t_i and held over the next interval.

Definition: integrate a simple strategy by summing its gains

Itô integral of a simple predictable process

For

$$\phi_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{(t_i, t_{i+1}]}(t),$$

define

$$\boxed{\int_0^T \phi_t \, \mathrm{d}W_t := \sum_{i=0}^{n-1} \xi_i \big(W_{t_{i+1}} - W_{t_i} \big).}$$

This is exactly the discrete self-financing gain associated with the strategy ϕ.

Units

Position $\times$ price move $=$ gain. The notation records a limiting trading operation, not an ordinary product of ϕ_t, $\mathrm{d}W_t$, and $\mathrm{d}t$.

Three sanity checks should become automatic

For $0 \leq a < b \leq T$ and a constant c,

$$\int_0^T c \, \mathrm{d}W_t = cW_T,$$

$$\int_0^T \mathbf{1}_{(a,b]}(t) \, \mathrm{d}W_t = W_b - W_a,$$

and, for $0 < u < T$,

$$\int_0^T \phi_t \, \mathrm{d}W_t = \int_0^u \phi_t \, \mathrm{d}W_t + \int_u^T \phi_t \, \mathrm{d}W_t.$$

Interpretation

The integral adds the Brownian moves on the intervals where the position is active, weighted by the position held there.

Worked example: a deterministic step strategy

Let $T = 3$ and

$$\phi_t = 2\mathbf{1}_{(0,1]}(t) - \mathbf{1}_{(1,3]}(t).$$

Then

$$I = \int_0^3 \phi_t \, \mathrm{d}W_t = 2W_1 - (W_3 - W_1).$$

The two increments are independent, so

$$I \sim \mathcal{N}(0, 4 \cdot 1 + 1 \cdot 2) = \mathcal{N}(0, 6).$$

Reading the variance

Each interval contributes (position)$^2 \times$ (length): $2^2 \cdot 1 + (-1)^2 \cdot 2 = 6$.

Exercise: switch the strategy after observing W_1

Let $A = \{W_1 > 0\}$ and define

$$\phi_t = \mathbf{1}_A \mathbf{1}_{(1,2]}(t).$$

Your turn – 3 minutes

1. Explain why ϕ is predictable.
2. Write $I = \int_0^2 \phi_t \, \mathrm{d}W_t$ without integral notation.
3. Compute $\mathbb{E}[I]$ and $\mathbb{E}[I^2]$.
4. Is I Gaussian?

This example separates two ideas: the integrand is random, but the next increment remains fresh.

Exercise debrief: a random integrand need not produce a Gaussian

Because $A \in \mathcal{F}_1$, the position is known before the interval $(1, 2]$. By definition,

$$I = \mathbf{1}_A(W_2 - W_1).$$

The event A is independent of $W_2 - W_1 \sim \mathcal{N}(0, 1)$, hence

$$\mathbb{E}[I] = 0, \qquad \mathbb{E}[I^2] = \mathbb{P}(A)\mathbb{E}[(W_2 - W_1)^2] = \tfrac{1}{2}.$$

But

$$\mathbb{P}(I = 0) = \mathbb{P}(A^c) = \tfrac{1}{2}.$$

Conclusion

The law is the mixture $\frac{1}{2}\delta_0 + \frac{1}{2}\mathcal{N}(0, 1)$, not a Gaussian law. Gaussianity is automatic for deterministic integrands, not for arbitrary random ones.

Linearity and time additivity mirror ordinary gains

For simple predictable ϕ, ψ and constants a, b,

$$\int_0^T (a\phi_t + b\psi_t)\,\mathrm{d}W_t = a\int_0^T \phi_t\,\mathrm{d}W_t + b\int_0^T \psi_t\,\mathrm{d}W_t.$$

For $0 \le s \le u \le t$,

$$\int_s^t \phi_r\,\mathrm{d}W_r = \int_s^u \phi_r\,\mathrm{d}W_r + \int_u^t \phi_r\,\mathrm{d}W_r.$$

If $\phi = \psi$ almost everywhere in $\Omega \times [0, T]$, their integrals are the same almost surely.

Practical message

Split time into convenient pieces, simplify the integrand on each piece, and then add the gains.

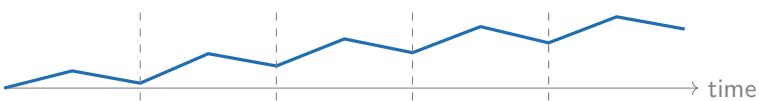
The integral up to time t is the accumulated gain process

For a simple ϕ, define

$$M_t = \int_0^t \phi_s \, \mathrm{d}W_s.$$

On an interval $(t_i, t_{i+1}]$,

$$M_t = M_{t_i} + \xi_i (W_t - W_{t_i}).$$



fresh Brownian gain on each holding interval

The continuous-time integral will preserve this interpretation after approximation.

The Ito Isometry Does the Construction

Disjoint Brownian increments are orthogonal

Let $\Delta W_i = W_{t_{i+1}} - W_{t_i}$.

Recall that Brownian increments over disjoint intervals are independent and $\Delta W_i \sim \mathcal{N}(0, t_{i+1} - t_i)$.

Your turn

Using these properties, compute:

1. For $i \neq j$,
$$\mathbb{E}[\Delta W_i \Delta W_j] = ?$$
2. What about
$$\mathbb{E}[(\Delta W_i)^2] = ?$$
3. If ξ_i is known at time t_i, what is
$$\mathbb{E}[\xi_i \Delta W_i \mid \mathcal{F}_{t_i}] = ?$$

Think about it

Which properties of Brownian motion make the cross terms disappear?

Disjoint Brownian increments are orthogonal

Let $\Delta W_i = W_{t_{i+1}} - W_{t_i}$. For $i \neq j$,

$$\mathbb{E}[\Delta W_i \Delta W_j] = 0,$$

while

$$\mathbb{E}[(\Delta W_i)^2] = t_{i+1} - t_i.$$

For predictable random coefficients, the useful conditional form is

$$\mathbb{E}[\xi_i \Delta W_i \mid \mathcal{F}_{t_i}] = 0.$$

Why cross terms disappear

When the later increment is averaged conditional on everything known before it, its conditional mean is zero.

This is the stochastic analogue of orthogonality of perpendicular vectors.

The Ito isometry converts variance into a time integral

Itô isometry for simple predictable processes

If ϕ is simple and predictable, then

$$\boxed{\mathbb{E}\left[\left(\int_0^T \phi_t \, \mathrm{d}W_t\right)^2\right] = \mathbb{E}\int_0^T \phi_t^2 \, \mathrm{d}t.}$$

In norm notation,

$$\left\| \int_0^T \phi_t \, \mathrm{d}W_t \right\|_{L^2(\Omega)} = \|\phi\|_{L^2(\Omega \times [0,T])}.$$

Meaning

The map $\phi \mapsto \int \phi \, \mathrm{d}W$ preserves squared length. That is exactly the stability needed to take limits.

The proof is one diagonal calculation

Write $I = \sum_i \xi_i \Delta W_i$. Then

$$\mathbb{E}[I^2] = \sum_i \mathbb{E}[\xi_i^2 (\Delta W_i)^2] + 2 \sum_{i<j} \mathbb{E}[\xi_i \Delta W_i \, \xi_j \Delta W_j].$$

For $i < j$, condition on $\mathcal{F}_{t_j}$: the second factor ΔW_j has conditional mean zero, so the cross term vanishes. On the diagonal,

$$\mathbb{E}[\xi_i^2 (\Delta W_i)^2] = \mathbb{E}[\xi_i^2](t_{i+1} - t_i).$$

Therefore

$$\mathbb{E}[I^2] = \sum_i \mathbb{E}[\xi_i^2](t_{i+1} - t_i) = \mathbb{E} \int_0^T \phi_t^2 \, \mathrm{d}t.$$

Mean and covariance come from the same computation

For simple predictable ϕ, ψ,

$$\mathbb{E}\left[\int_0^T \phi_t \, \mathrm{d}W_t\right] = 0$$

and, by polarization,

$$\boxed{\mathbb{E}\left[\left(\int_0^T \phi_t \, \mathrm{d}W_t\right)\left(\int_0^T \psi_t \, \mathrm{d}W_t\right)\right] = \mathbb{E}\int_0^T \phi_t \psi_t \, \mathrm{d}t.}$$

Orthogonality criterion

If $\phi_t \psi_t = 0$ almost everywhere, the two integrals are uncorrelated. For deterministic integrands they are jointly Gaussian, hence independent.

Energy of the strategy equals variance of the gain

Define the pathwise exposure budget and its average energy by

$$\mathcal{E}_T(\phi) = \int_0^T \phi_t^2 \, \mathrm{d}t, \qquad \mathbb{E}[\mathcal{E}_T(\phi)] = \mathbb{E} \int_0^T \phi_t^2 \, \mathrm{d}t.$$

- Doubling ϕ multiplies gain variance by 4.
- Holding the same position twice as long doubles variance.
- Moving exposure across time can preserve terminal variance while changing the gain path.

A simple risk bound

If $|\phi_t| \leq L$, then

$$\mathrm{Var}\left(\int_0^T \phi_t \, \mathrm{d}W_t \right) \leq L^2 T.$$

Exercise: two strategies with the same terminal risk

On $[0,1]$, compare

$$\phi_t = 1, \qquad \psi_t = \sqrt{2}\,\mathbf{1}_{(0,1/2]}(t).$$

Your turn – 2 minutes

1. Compute the energy of each strategy.
2. Identify the two stochastic integrals.
3. Compare their terminal distributions.
4. Are the gain processes identical before time 1?

The isometry compares terminal risk, not the timing of that risk.

Exercise debrief: equal energy gives equal Gaussian laws here

Both deterministic strategies have unit energy:

$$\int_0^1 \phi_t^2 \, \mathrm{d}t = 1, \qquad \int_0^1 \psi_t^2 \, \mathrm{d}t = 2 \cdot \tfrac{1}{2} = 1.$$

Their gains are

$$\int_0^1 \phi_t \, \mathrm{d}W_t = W_1, \qquad \int_0^1 \psi_t \, \mathrm{d}W_t = \sqrt{2}\, W_{1/2}.$$

Thus both are $\mathcal{N}(0,1)$.

But the paths differ

The second strategy stops accumulating risk after time $1/2$; the first remains exposed until time 1. Same terminal distribution does not mean same gain process.

The completion idea is the entire extension argument

Suppose simple predictable processes ϕ^n approximate ϕ so that

$$\mathbb{E}\int_0^T |\phi_t^n - \phi_t|^2 \,\mathrm{d}t \longrightarrow 0.$$

By the isometry,

$$\mathbb{E}\left[\left|\int_0^T \phi_t^n \,\mathrm{d}W_t - \int_0^T \phi_t^m \,\mathrm{d}W_t\right|^2\right] = \mathbb{E}\int_0^T |\phi_t^n - \phi_t^m|^2 \,\mathrm{d}t.$$

The integrals are therefore Cauchy in $L^2(\Omega)$, a complete space.

Construction in one sentence

Approximate the strategy, integrate the simple approximations, and take the L^2 limit.

Definition for a general square-integrable predictable integrand

Let ϕ be predictable with

$$\mathbb{E}\int_0^T \phi_t^2 \,\mathrm{d}t < \infty.$$

Choose simple predictable ϕ^n converging to ϕ in $L^2(\Omega \times [0,T])$.

General Itô integral

$$\boxed{\int_0^T \phi_t \,\mathrm{d}W_t := L^2\text{-}\lim_{n\to\infty} \int_0^T \phi_t^n \,\mathrm{d}W_t.}$$

The isometry shows that the limit exists and is independent of the chosen approximation. Linearity, mean zero, and the isometry pass to the limit.

The Integral Is a Martingale Gain Process

Before saying "martingale," check integrability

Let ϕ be predictable and assume, for every finite T,

$$\mathbb{E}\int_0^T \phi_s^2 \,\mathrm{d}s < \infty.$$

Then, for $M_t = \int_0^t \phi_s \,\mathrm{d}W_s$, the isometry and Cauchy–Schwarz give

$$\mathbb{E}|M_t| \le \left(\mathbb{E}[M_t^2]\right)^{1/2} = \left(\mathbb{E}\int_0^t \phi_s^2 \,\mathrm{d}s\right)^{1/2} < \infty.$$

Thus $M_t \in L^2 \subset L^1$: its conditional expectation is well defined.

Do not confuse two levels of assumptions

If only $\int_0^T \phi_s^2 \,\mathrm{d}s < \infty$ almost surely, the integral is generally a *local* martingale. The expectation condition above gives the square-integrable, hence true, martingale used here.

Finite expected energy gives a true martingale

For a predictable ϕ satisfying

$$\mathbb{E}\int_0^T \phi_s^2 \, \mathrm{d}s < \infty, \qquad M_t = \int_0^t \phi_s \, \mathrm{d}W_s,$$

Square-integrable martingale property

The process M is adapted, $M_t \in L^2$, and, for $0 \le s \le t \le T$,

$$\boxed{\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s.}$$

Equivalently,

$$\mathbb{E}\left[\int_s^t \phi_u \, \mathrm{d}W_u \,\middle|\, \mathcal{F}_s\right] = 0.$$

The future gain may have state-dependent risk, but its conditional mean is zero.

The martingale proof repeats the discrete no-drift argument

For a simple process on a grid containing s, split

$$M_t = M_s + \sum_{t_i \geq s} \xi_i \Delta W_i.$$

For each future increment,

$$\mathbb{E}[\xi_i \Delta W_i \mid \mathcal{F}_{t_i}] = \xi_i \mathbb{E}[\Delta W_i \mid \mathcal{F}_{t_i}] = 0.$$

Iterated conditioning gives

$$\mathbb{E}[M_t - M_s \mid \mathcal{F}_s] = 0.$$

The general case follows by L^2 approximation. Since L^2 convergence implies L^1 convergence, conditional expectations pass to the limit.

Proof pattern to remember

Predictable coefficient $\times$ fresh centered increment $\Rightarrow$ zero conditional mean.

Conditional variance measures the risk still ahead

The conditional version of the isometry gives

$$\boxed{\mathbb{E}\left[\left(\int_s^t \phi_u \,\mathrm{d}W_u\right)^2 \middle| \mathcal{F}_s\right] = \mathbb{E}\left[\int_s^t \phi_u^2 \,\mathrm{d}u \middle| \mathcal{F}_s\right].}$$

Conditional mean

The future stochastic gain is zero on average given current information.

Conditional variance

Current information can change how much future variance the strategy will accumulate.

This distinction is central in hedging: no predictable drift does not mean no risk.

Exercise: a state-dependent position changes future variance

On $[0,2]$, define

$$\phi_t = \mathbf{1}_{(0,1]}(t) + \big(1 + \mathbf{1}_{\{W_1>0\}}\big)\mathbf{1}_{(1,2]}(t), \qquad M_t = \int_0^t \phi_s \, \mathrm{d}W_s.$$

Your turn – 3 minutes

Compute, in terms of W_1,

1. M_1;
2. $\mathbb{E}[M_2 \mid \mathcal{F}_1]$;
3. $\mathrm{Var}(M_2 - M_1 \mid \mathcal{F}_1)$.

The position doubles after time 1 precisely on the event $\{W_1 > 0\}$.

Exercise debrief: the forecast is unchanged, the risk is not

Up to time 1, the position is one, so

$$M_1 = W_1.$$

The martingale property yields

$$\mathbb{E}[M_2 \mid \mathcal{F}_1] = M_1 = W_1.$$

Over $(1, 2]$, the position is already known at time 1, hence

$$\mathrm{Var}(M_2 - M_1 \mid \mathcal{F}_1) = \left(1 + \mathbf{1}_{\{W_1 > 0\}}\right)^2 = \begin{cases} 1, & W_1 \le 0, \\ 4, & W_1 > 0. \end{cases}$$

Takeaway

Adaptation can make future variance depend on the current state without creating conditional drift.

Quadratic variation is the integral's variance clock

For

$$M_t = \int_0^t \phi_s \, \mathrm{d}W_s,$$

the quadratic variation satisfies

$$\boxed{[M]_t = \int_0^t \phi_s^2 \, \mathrm{d}s.}$$

For a simple strategy, this is suggested directly by

$$\sum_i (\Delta M_i)^2 = \sum_i \xi_i^2 (\Delta W_i)^2 \approx \sum_i \xi_i^2 \Delta t_i.$$

Interpretation

Calendar time runs at unit speed for W. The gain process runs on the random clock generated by squared exposure.

Stopping a strategy means switching the integrand off

Let τ be a stopping time. Switching the position off at τ gives

$$\boxed{\int_0^t \mathbf{1}_{\{s \le \tau\}} \, \mathrm{d}W_s = W_{t \wedge \tau}.}$$

For each finite t, the stopped integrand is square-integrable because

$$\mathbb{E} \int_0^t \mathbf{1}^2_{\{s \le \tau\}} \, \mathrm{d}s = \mathbb{E}[t \wedge \tau] \le t.$$

The martingale property and the isometry therefore apply:

$$\boxed{\mathbb{E}[W_{t \wedge \tau}] = 0, \qquad \mathbb{E}[W^2_{t \wedge \tau}] = \mathbb{E}[t \wedge \tau].}$$

This check is what licenses the two expectation identities; stopping alone is not a substitute for integrability.

Application: stop Brownian exposure at a barrier

Let $\tau_a = \inf\{t \geq 0 : W_t = a\}$, with $a > 0$. The stopped exposure

$$M_t = \int_0^t \mathbf{1}_{\{s \leq \tau_a\}} \, \mathrm{d}W_s = W_{t \wedge \tau_a}$$

never continues after the upper barrier is reached.

Your turn

Use only the martingale property and isometry to identify

$$\mathbb{E}[M_t] \qquad \text{and} \qquad \mathbb{E}[M_t^2].$$

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Your turn

Use only the martingale property and isometry to identify

$$\mathbb{E}[M_t] \qquad \text{and} \qquad \mathbb{E}[M_t^2].$$

Answer

$$\mathbb{E}[M_t] = 0, \qquad \mathbb{E}[M_t^2] = \mathbb{E}[t \wedge \tau_a].$$

No density of τ_a is needed.

Predictable Brownian trading transforms risk, not drift

For every predictable strategy satisfying $\mathbb{E}\int_0^T \phi_t^2 \, \mathrm{d}t < \infty$,

$$\mathbb{E}\left[\int_0^T \phi_t \, \mathrm{d}W_t\right] = 0.$$

Yet the distribution can be:

- Gaussian when ϕ is deterministic;
- a non-Gaussian mixture when ϕ reacts to past states;
- stopped, skewed, or state-dependent for more complex strategies.

What remains invariant

The gain process is a square-integrable martingale and its second moment is exactly the expected integrand energy.

This is the structural content of stochastic integration before any Itô formula is used.

Deterministic Integrands Give Explicit Gaussian Laws

A deterministic integrand produces a Gaussian random variable

Let $f \in L^2(0, T)$ be deterministic. Then

$$I_f = \int_0^T f(t)\,\mathrm{d}W_t$$

is a limit of Gaussian linear combinations of Brownian increments.

Distribution

$$\boxed{I_f \sim \mathcal{N}\left(0, \int_0^T f(t)^2\,\mathrm{d}t\right).}$$

The mean is zero and the variance comes from the isometry.

Why the Gaussian law survives the limit

The characteristic functions converge to that of the centered Gaussian with the limiting variance.

Inner products of functions become covariances of integrals

For deterministic $f, g \in L^2(0, T)$,

$$\operatorname{Cov}\left(\int_0^T f \,\mathrm{d}W, \int_0^T g \,\mathrm{d}W\right) = \int_0^T f(t)g(t)\,\mathrm{d}t.$$

Moreover, the pair is jointly Gaussian.

Independence criterion

If

$$\int_0^T f(t)g(t)\,\mathrm{d}t = 0,$$

then $\int f \,\mathrm{d}W$ and $\int g \,\mathrm{d}W$ are independent.

Orthogonal deterministic exposure profiles generate independent Gaussian risks.

Worked application: a linearly increasing exposure

Take $f(t) = t$. Then

$$I_T = \int_0^T t \, \mathrm{d}W_t$$

is centered Gaussian with

$$\mathrm{Var}(I_T) = \int_0^T t^2 \, \mathrm{d}t = \frac{T^3}{3}.$$

Therefore

$$\boxed{I_T \sim \mathcal{N}\left(0, \frac{T^3}{3}\right).}$$

Scaling check

Brownian noise contributes $\sqrt{\mathrm{d}t}$, while the position grows like t. The resulting standard deviation scales like $T^{3/2}$.

Exercise: condition an integral on the terminal Brownian value

Let

$$I_T = \int_0^T t \, \mathrm{d}W_t.$$

Your turn – 4 minutes

1. Write W_T as a stochastic integral.
2. Compute $\mathrm{Cov}(I_T, W_T)$.
3. Use Gaussian regression to find $\mathbb{E}[I_T \mid W_T]$.
4. Compute $\mathrm{Var}(I_T \mid W_T)$.

Recall that for a centered Gaussian pair (X, Y),

$$\mathbb{E}[X \mid Y] = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(Y)} Y.$$

Exercise debrief: conditioning is projection in function space

Since

$$W_T = \int_0^T 1 \, \mathrm{d}W_t,$$

the covariance formula gives

$$\mathrm{Cov}(I_T, W_T) = \int_0^T t \, \mathrm{d}t = \frac{T^2}{2}.$$

Because $\mathrm{Var}(W_T) = T$,

$$\boxed{\mathbb{E}[I_T \mid W_T] = \frac{T}{2} W_T.}$$

The residual conditional variance is

$$\mathrm{Var}(I_T \mid W_T) = \frac{T^3}{3} - \frac{(T^2/2)^2}{T} = \frac{T^3}{12}.$$

Only the component of the function $t \mapsto t$ parallel to the constant function is revealed by W_T.

Exercise: construct a Gaussian risk independent of W_T

We seek a nonzero deterministic g such that

$$Y = \int_0^T g(t)\,\mathrm{d}W_t$$

is independent of W_T.

Your turn – 2 minutes

1. Translate independence into an integral condition on g.
2. Find a simple affine function $g(t) = t - c$ satisfying it.
3. Compute $\mathrm{Var}(Y)$.

The problem is ordinary orthogonality in $L^2(0, T)$.

Exercise debrief: center the exposure profile

Independence from $W_T = \int_0^T 1 \, \mathrm{d}W_t$ requires

$$\int_0^T g(t) \, \mathrm{d}t = 0.$$

Choose

$$g(t) = t - \frac{T}{2}.$$

Then

$$\mathrm{Cov}(Y, W_T) = 0,$$

and joint Gaussianity gives $Y \perp\!\!\!\perp W_T$. Its variance is

$$\mathrm{Var}(Y) = \int_0^T \left(t - \frac{T}{2} \right)^2 \mathrm{d}t = \frac{T^3}{12}.$$

Geometric reading

Subtracting the average removes the component aligned with the terminal Brownian shock.

Deterministic integrands create time-changed Brownian motions

Define

$$M_t = \int_0^t f(s)\,\mathrm{d}W_s, \qquad q(t) = \int_0^t f(s)^2\,\mathrm{d}s.$$

The process M is centered Gaussian, has independent increments, and

$$\mathrm{Var}(M_t - M_s) = q(t) - q(s).$$

Time-change viewpoint

$$(M_t)_{t\geq 0} \overset{\text{law}}{=} (B_{q(t)})_{t\geq 0}$$

for a Brownian motion B.

For $f(t) = t$, the stochastic clock is $q(t) = t^3/3$.

Integration by parts with a deterministic integrand

Sufficient regularity

Assume that $f : [0, T] \to \mathbb{R}$ is deterministic and C^1. More generally, it is enough that f be absolutely continuous with $f' \in L^2(0, T)$. Then $f \in L^2(0, T)$, and both integrals below exist.

Discrete summation by parts gives, in the limit,

$$f(T)W_T - f(0)W_0 = \int_0^T f(t)\,\mathrm{d}W_t + \int_0^T W_t f'(t)\,\mathrm{d}t.$$

Since $W_0 = 0$,

$$\boxed{\int_0^T f(t)\,\mathrm{d}W_t = f(T)W_T - \int_0^T W_t f'(t)\,\mathrm{d}t.}$$

No Itô formula is needed

The deterministic path has finite variation, so its quadratic covariation with W is zero: there is no correction term.

Application: identify the law of the Brownian area

Set

$$A_T = \int_0^T W_t \, \mathrm{d}t.$$

Using integration by parts with $f(t) = t$,

$$A_T = TW_T - \int_0^T t \, \mathrm{d}W_t.$$

Thus A_T is Gaussian and centered. From the covariance calculations,

$$\mathrm{Var}(A_T) = T^2 \, \mathrm{Var}(W_T) + \mathrm{Var}(I_T) - 2T \, \mathrm{Cov}(W_T, I_T) = \frac{T^3}{3}.$$

Therefore

$$\boxed{A_T \sim \mathcal{N}\left(0, \frac{T^3}{3}\right).}$$

This path-dependent functional is handled without evaluating any density by hand.

Left, Right, and Midpoint Conventions

Three natural sums lead to three different limits

For $\phi_t = W_t$ and partition π, define

$$L^\pi = \sum_i W_{t_i} \Delta W_i, \qquad M^\pi = \sum_i \frac{W_{t_i} + W_{t_{i+1}}}{2} \Delta W_i, \qquad R^\pi = \sum_i W_{t_{i+1}} \Delta W_i.$$

Sum	Information	Limiting convention
left	known before the move	Itô
midpoint	uses both endpoints	Stratonovich
right	known after the move	anticipative here

Brownian quadratic variation prevents these sums from collapsing to the same answer.

An exact algebraic identity isolates the correction

For every interval,

$$W_{t_{i+1}}^2 - W_{t_i}^2 = 2W_{t_i}\Delta W_i + (\Delta W_i)^2.$$

Summing over i,

$$W_T^2 = 2L^\pi + \sum_i (\Delta W_i)^2.$$

As the mesh tends to zero,

$$\sum_i (\Delta W_i)^2 \longrightarrow T.$$

This is the whole mechanism

The square of a Brownian increment is small on one interval but accumulates to an order-one correction over many intervals.

The first nontrivial stochastic integral can be computed directly

Passing to the limit in the left-sum identity gives

$$\boxed{\int_0^T W_t \,\mathrm{d}W_t = \frac{1}{2}(W_T^2 - T).}$$

Compare this with the classical formula $\int x \,\mathrm{d}x = \frac{1}{2}x^2$.

Ito correction

The additional $-T/2$ exactly compensates for Brownian quadratic variation and restores the zero-mean martingale property:

$$\mathbb{E}\left[\frac{1}{2}(W_T^2 - T)\right] = 0.$$

No general Itô formula was needed – only a telescoping identity and $[W]_T = T$.

Right and midpoint sums reveal the size of the convention effect

Since

$$R^{\pi} = L^{\pi} + \sum_i (\Delta W_i)^2,$$

the right-point limit is

$$\frac{1}{2}(W_T^2 - T) + T = \frac{1}{2}(W_T^2 + T).$$

The midpoint sum is the average of left and right sums, so its limit is

$$\frac{1}{2}W_T^2.$$

Convention	Limit for $\int_0^T W \,\mathrm{d}W$
left / Itô	$\frac{1}{2}(W_T^2 - T)$
midpoint / Stratonovich	$\frac{1}{2}W_T^2$
right	$\frac{1}{2}(W_T^2 + T)$

Exercise: verify the isometry on $\int_0^T W_t \, \mathrm{d}W_t$

Let

$$I_T = \int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2}(W_T^2 - T).$$

Your turn – 3 minutes

1. Compute $\mathbb{E}[I_T]$.
2. Using $\mathbb{E}[W_T^4] = 3T^2$, compute $\mathbb{E}[I_T^2]$.
3. Check that the result agrees with the Itô isometry.

This is a useful consistency test between the explicit formula and the abstract construction.

Exercise debrief: the explicit formula matches the isometry

Since $\mathbb{E}[W_T^2] = T$,

$$\mathbb{E}[I_T] = 0.$$

Moreover,

$$\mathbb{E}[I_T^2] = \frac{1}{4}\mathbb{E}[(W_T^2 - T)^2] = \frac{1}{4}(3T^2 - 2T^2 + T^2) = \frac{T^2}{2}.$$

The isometry gives the same answer:

$$\mathbb{E}\left[\left(\int_0^T W_t \,\mathrm{d}W_t\right)^2\right] = \mathbb{E}\int_0^T W_t^2 \,\mathrm{d}t = \int_0^T t \,\mathrm{d}t = \frac{T^2}{2}.$$

Two routes, one answer.

Seeing the next increment would create a fake arbitrage

Suppose illegally that the trader could choose

$$\phi_{t_i} = \text{sign}(\Delta W_i)$$

before reporting the gain. Then

$$\sum_i \phi_{t_i} \Delta W_i = \sum_i |\Delta W_i| \geq 0,$$

and it is strictly positive with probability one on a nontrivial grid.

What failed?

The position is not $\mathcal{F}_{t_i}$-measurable: it uses the very move it claims to predict. This is not a trading strategy; it is hindsight.

Predictability is therefore both the mathematical and economic boundary of the Itô integral.

Simulation means implementing the left-point rule

On the uniform grid $t_i = i\Delta t$, with $\Delta t = T/n$, generate

$$Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad \Delta W_i = \sqrt{\Delta t}\, Z_i.$$

Starting from $W_0 = I_0 = 0$, repeat at each step:

1. compute ϕ_i using only information available at t_i;
2. update $I_{i+1} = I_i + \phi_i \Delta W_i$;
3. update $W_{t_{i+1}} = W_{t_i} + \Delta W_i$.

Then

$$I_n = \sum_{i=0}^{n-1} \phi_i \Delta W_i \approx \int_0^T \phi_t \, \mathrm{d}W_t.$$

The order is mathematical, not cosmetic

If ϕ_i uses $W_{t_{i+1}}$ or the newly generated ΔW_i, the scheme is no longer a left-point Itô sum.

A theoretical benchmark for the simulation error

For the integral $I = \int_0^T W_t \, \mathrm{d}W_t$, simulate the left sum

$$I^{(n)} = \sum_{i=0}^{n-1} W_{t_i} \Delta W_i.$$

The exact identity $I = \frac{1}{2}(W_T^2 - T)$ gives

$$I - I^{(n)} = \frac{1}{2} \sum_{i=0}^{n-1} \big((\Delta W_i)^2 - \Delta t \big).$$

On a uniform grid, independence and $\mathrm{Var}((\Delta W_i)^2) = 2(\Delta t)^2$ yield

$$\boxed{\mathbb{E}\big[(I - I^{(n)})^2\big] = \frac{T^2}{2n}.}$$

A useful unit test

The root-mean-square error should decay like $n^{-1/2}$. This tests the simulation against a theorem, not a visually plausible path.

Classic simulation traps and how to detect them

- **Wrong Brownian scale:** using $\Delta t\, Z_i$ instead of $\sqrt{\Delta t}\, Z_i$ destroys the variance clock.
- **Looking one step ahead:** computing ϕ_i from $W_{t_{i+1}}$ produces a right-point, anticipative sum and typically adds a bias.
- **Wrong noise coupling:** if the integrand and the integral are driven by the same Brownian motion, they must use the same increments; fresh independent noise changes their joint law.
- **Confusing $\mathrm{d}W$ and $\mathrm{d}t$:** $\sum_i \phi_i \Delta W_i$ approximates a stochastic integral, while $\sum_i \phi_i \Delta t$ approximates an ordinary time integral.
- **Checking only the mean:** also compare the empirical second moment with $\mathbb{E}\int_0^T \phi_t^2\, \mathrm{d}t$, and test several grid sizes.

Filtration check

At every step, ask: was ϕ_i fully determined before ΔW_i was used?

What to retain before learning the Ito formula

1. $\int \phi \, \mathrm{d}W$ is the limit of left-point, non-anticipative gains built first for simple predictable strategies.
2. The isometry converts integrand energy into gain variance and makes the L^2 extension possible.
3. Under $\mathbb{E} \int_0^T \phi_t^2 \, \mathrm{d}t < \infty$, the integral process is a square-integrable martingale with quadratic variation $\int \phi^2 \, \mathrm{d}t$.
4. Deterministic integrands give explicit jointly Gaussian variables.
5. Brownian quadratic variation creates the correction

$$\int_0^T W_t \, \mathrm{d}W_t = \tfrac{1}{2}(W_T^2 - T).$$

6. Simulation uses the same left-point information constraint and is checked through the isometry or an exact benchmark.

Next lecture

The Itô formula turns this correction mechanism into a general chain rule.